

Лабораторная работа №13

Фильтр пакетов (firewalld)

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Цель работы

Получить практические навыки настройки брандмауэра в Linux с помощью `firewall-cmd` и `firewall-config`.

Ход выполнения

Определение активной зоны и доступных сервисов

```
amnauruzova@amnauruzova:~$ su
Password:
root@amnauruzova:/home/amnauruzova# firewall-cmd --get-default-zone
public
root@amnauruzova:/home/amnauruzova# firewall-cmd --get-zones
block dmz drop external home internal nm-shared public trusted work
root@amnauruzova:/home/amnauruzova# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 anno-1800 apcupsd
aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-r
pc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civilization-iv
civilization-v cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client dis
tcc dns dns-over-quic dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server fac
torio finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gang
lia-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps iperf2 iperf3 ipfs ipp
ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-a
pi kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kub
e-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps l
ibvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft mi
nidlna mndp mongodb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios
-ns netdata-dashboard nfs nfs3 nmea-0183 nripe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmco
nsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp p
s2link ps3netshr ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rs
yncd rtp salt-master samba samba-client samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submiss
ion smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh statsrv steam-lan-transfer ste
am-streaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing
-relay synergy syscomlan syslog syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn t
urns upnp-client vdsim vnc-server vrrp warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-disco
very-host ws-discovery-tcp ws-discovery-udp wsdd wsdd-http wsmans wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-serve
r zabbix-agent zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-services
cockpit dhcpv6-client ssh
root@amnauruzova:/home/amnauruzova# █
```

Рис. 1: Определение зоны и сервисов

Детальный вывод конфигурации зоны

```
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all --zone=public
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@amnauruzova:/home/amnauruzova#
```

Добавление службы VNC (runtime)

```
root@amnauruzova:/home/amnauruzova# firewall-cmd --add-service=vnc-server
success
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@amnauruzova:/home/amnauruzova# systemctl restart firewalld.service
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
```

Добавление VNC в постоянную конфигурацию

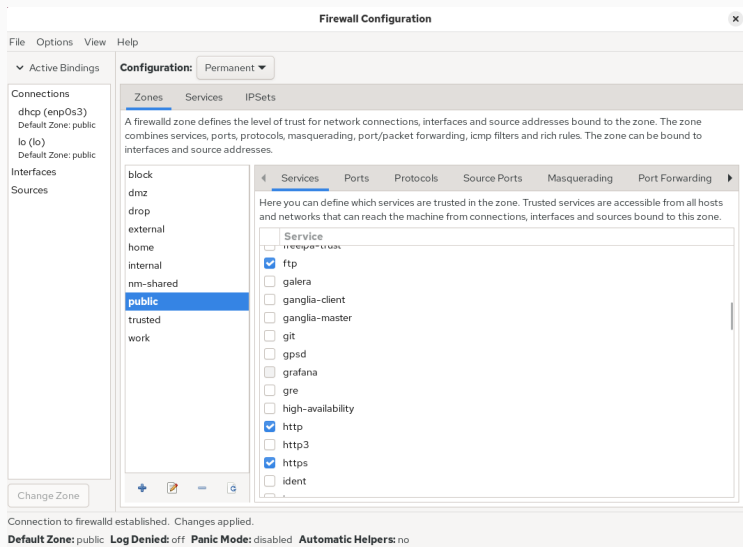
```
root@amnauruzova:/home/amnauruzova# firewall-cmd --add-service=vnc-server --permanent
success
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@amnauruzova:/home/amnauruzova# firewall-cmd --reload
success
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ssh vnc-server
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
```

Открытие порта 2022/TCP

```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# firewall-cmd --add-port=2022/tcp --permanent  
success  
root@amnauruzova:/home/amnauruzova# firewall-cmd --reload  
success  
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all  
public (default, active)  
  target: default  
  ingress-priority: 0  
  egress-priority: 0  
  icmp-block-inversion: no  
  interfaces: enp0s3  
  sources:  
  services: cockpit dhcpv6-client ssh vnc-server  
  ports: 2022/tcp  
  protocols:  
  forward: yes  
  masquerade: no  
  forward-ports:  
  source-ports:  
  icmp-blocks:  
  rich rules:  
root@amnauruzova:/home/amnauruzova#
```

Рис. 5: Добавление порта

firewall-config — включение сервисов



Добавление порта в GUI

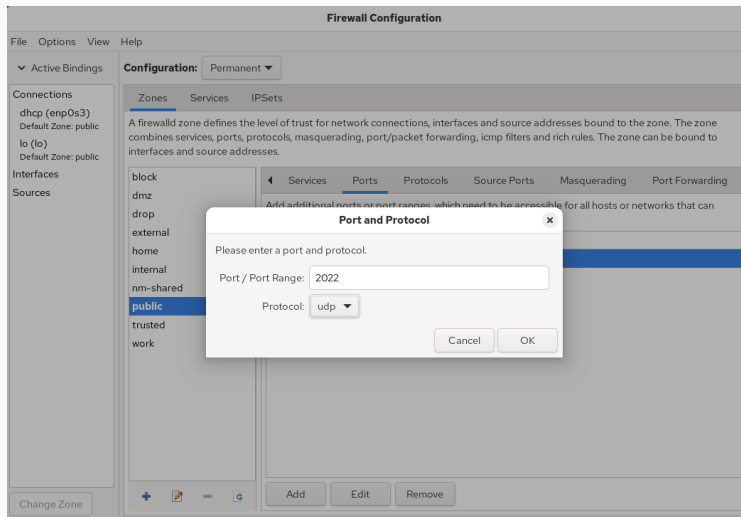


Рис. 7: Добавление UDP порта

Применение конфигурации

```
root@amnauruzova:/home/amnauruzova#  
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all  
public (default, active)  
  target: default  
  ingress-priority: 0  
  egress-priority: 0  
  icmp-block-inversion: no  
  interfaces: enp0s3  
  sources:  
  services: cockpit dhcpv6-client ssh vnc-server  
  ports: 2022/tcp  
  protocols:  
  forward: yes  
  masquerade: no  
  forward-ports:  
  source-ports:  
  icmp-blocks:  
  rich rules:  
root@amnauruzova:/home/amnauruzova# firewall-cmd --reload  
success  
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all  
public (default, active)  
  target: default  
  ingress-priority: 0  
  egress-priority: 0  
  icmp-block-inversion: no  
  interfaces: enp0s3  
  sources:  
  services: cockpit dhcpv6-client ftp http https ssh vnc-server  
  ports: 2022/tcp 2022/udp  
  protocols:  
  forward: yes  
  masquerade: no  
  forward-ports:
```

```
root@amnauruzova:/home/amnauruzova# firewall-cmd --reload
success
root@amnauruzova:/home/amnauruzova# firewall-cmd --list-all
public (default, active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp0s3
  sources:
  services: cockpit dhcpv6-client ftp http https imap pop3 smtp ssh telnet vnc-server
  ports: 2022/tcp 2022/udp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
root@amnauruzova:/home/amnauruzova# █
```

Рис. 9: Финальная конфигурация

Итоги работы

- Освоена работа с `firewall-cmd` (CLI) и `firewall-config` (GUI)
- Изучены **runtime** и **permanent** конфигурации
- Выполнено добавление сервисов и портов
- Получены практические навыки управления сетевой безопасностью

Настройка брандмауэра позволяет контролировать сетевой доступ и повышать безопасность системы Linux.